Prashant K. Jha

Assistant Professor

Leslie A. Rose Department of Mechanical Engineering
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Research Profile

Principal Investigator of the Computational Engineering Analysis and Design (CEAD) Lab. My group develops predictive mechanics and reliable computational methods for complex materials and interfaces. Current work spans constitutive and interfacial mechanics of active soft materials, fracture and contact in particle-reinforced and multiparticle systems, and error-controlled hybrid data–mechanics methods for inference, design, and computational twins.

Academic Appointments

Assistant Professor
Aug 2024 – present
Department of Mechanical Engineering
South Dakota School of Mines & Technology, Rapid City, SD 57701, USA

Prior Appointments

Lecturer (Asst. Professor)
Nov 2023 – Aug 2024
School of Mechanical and Design Engineering
University of Portsmouth, Portsmouth PO1 2UP, UK

Research Affiliate
Nov 2023 – Oct 2024
Oden Institute for Computational Engineering and Sciences
The University of Texas at Austin, Austin, TX 78712, USA

Research Associate
Dec 2020 – Nov 2023
Oden Institute for Computational Engineering and Sciences
The University of Texas at Austin, Austin, TX 78712, USA
Advisor: Prof. J. Tinsley Oden

Peter O'Donnell Postdoctoral Fellow
Aug 2019 – Nov 2020
Oden Institute for Computational Engineering and Sciences
The University of Texas at Austin, Austin, TX 78712, USA
Advisor: Prof. J. Tinsley Oden

Postdoctoral Fellow
Oct 2016 – Jul 2019
Department of Mathematics
Louisiana State University, Baton Rouge, LA 70803, USA
Advisor: Prof. Robert Lipton

Education

Ph.D.
2012 – 2016
Civil and Environmental Engineering
Carnegie Mellon University, Pittsburgh, PA 15213, USA
Thesis: Coarse graining of electric field interactions with materials
Advisor: Prof. Kaushik Dayal

M.E.
2010 – 2012
Mechanical Engineering
Indian Institute of Science, Bengaluru, KA 560012, India
Thesis: A monolithic strategy for fluid-structure interaction in compressible flow
Advisor: Prof. Chandrashekhar S. Jog

B.E.
2006 – 2010
Mechanical Engineering
Govt. Engineering College, Raipur, CG 492001, India

Research Support

PI-led research support: $290,000. Prior collaborative support as co-PI: $50,000.

Funded Projects

NSF CMMI–ERI	**PI**, Adaptive Multi-Fidelity Framework for Modeling Heterogeneous Materials Under Extreme Conditions; $200,000; June 2025–May 2027.
SDBOR CRG	**PI**, Modeling and Design of Magnetic Soft Materials Accelerated by Neural Operators; $90,000; June 2025–April 2027.
MD Anderson	**Co-PI**, A Mechanistic Tumor Growth Model for Hyperpolarized MRI; $50,000; September 2020–March 2022. PI: David T. Fuentes.

Pending Proposals

NSF CAREER	**PI**, Mechanics of Contact and Adhesion in Nonlinear, Field-Responsive Soft Materials; submitted July 2026; under review.
NSF CDS&E	**PI**, Prediction-Specific Error Assessment and Control for Neural Operators in Nonlinear Mechanics; submitted September 2026; under review.

Selected Honors

Aug 2019	Peter O'Donnell Postdoctoral Fellowship, The University of Texas at Austin.
May 2013	Teaching Assistant Award, finite element methods, Carnegie Mellon University.
May 2010	All India Rank 31 (score: 957/1000), Graduate Aptitude Test in Engineering.

Publications [Google Scholar]

Published

1. P. K. **Jha** and K. Dayal, "Continuum limit of nonlocal electrostatics in random media," *Mathematics and Mechanics of Solids*, 2026.
2. P. K. **Jha**, "From theory to application: A practical introduction to neural operators in scientific computing," *Mathematics*, vol. 14, no. 13, 2026.
3. M. Nandyala, A. Lanham, P. K. **Jha**, C. Wu, J. D. Hazle, T. E. Yankeelov, J. Stafford, A. A. El-Gendy, and D. Fuentes, "An information-theoretic framework for optimal experimental design in magnetic nanoparticle hyperthermia," *Applied Mathematical Modelling*, p. 116861, 2026.
4. P. K. **Jha**, "PeriDEM – high-fidelity modeling of granular media consisting of deformable complex-shaped particles," *Journal of Open Source Software*, vol. 10, no. 116, p. 7525, 2025.
5. A. Patra, D. Faghihi, P. K. **Jha**, L. Cao, and K. A. Farrell-Maupin, "Predictive science: A quest for the holy," *Computing in Science & Engineering*, vol. 28, no. 1, pp. 30–40, 2026.
6. P. K. **Jha**, P. Diehl, and R. Lipton, "Nodal finite element approximation of peridynamics," *Computer Methods in Applied Mechanics and Engineering*, vol. 434, p. 117519, 2025.
7. P. K. **Jha**, "Residual-based error corrector operator to enhance accuracy and reliability of neural operator surrogates of nonlinear variational boundary-value problems," *Computer Methods in Applied Mechanics and Engineering*, vol. 419, p. 116595, 2024.
8. P. K. **Jha**, C. Walker, D. Mitchell, J. T. Oden, D. Schellingerhout, J. A. Bankson, and D. T. Fuentes, "Mutual-information based optimal experimental design for hyperpolarized 13C-pyruvate MRI," *Scientific Reports*, vol. 13, no. 1, p. 18047, 2023.
9. L. Cao, T. O'Leary-Roseberry, P. K. **Jha**, J. T. Oden, and O. Ghattas, "Residual-based error correction for neural operator accelerated infinite-dimensional Bayesian inverse problems," *Journal of Computational Physics*, p. 112104, 2023.
10. P. K. **Jha**, T. Breitzman, and K. Dayal, "Discrete-to-continuum limits of long-range electrical interactions in nanostructures," *Archive for Rational Mechanics and Analysis*, vol. 247, no. 2, p. 29, 2023.

11. P. K. **Jha**, J. Marshall, J. Knap, and K. Dayal, "Atomic-to-continuum multiscale modeling of defects in crystals with nonlocal electrostatic interactions," *Journal of Applied Mechanics*, vol. 90, no. 2, 2022.
12. P. K. **Jha** and J. T. Oden, "Goal-oriented a-posteriori estimation of model error as an aid to parameter estimation," *Journal of Computational Physics*, vol. 470, p. 111575, 2022.
13. P. K. **Jha** and P. Diehl, "NLMech: Implementation of finite difference/meshfree discretization of nonlocal fracture models," *Journal of Open Source Software*, vol. 6, no. 65, p. 3020, 2021.
14. P. Gadikar, P. Diehl, and P. K. **Jha**, "Load balancing for distributed nonlocal models within asynchronous many-task systems," in *2021 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW)*, (Los Alamitos, CA, USA), pp. 669–678, IEEE Computer Society, 2021.
15. D. A. Hormuth, C. M. Phillips, C. Wu, E. A. B. F. Lima, G. Lorenzo, P. K. **Jha**, A. M. Jarrett, J. T. Oden, and T. E. Yankeelov, "Biologically-based mathematical modeling of tumor vasculature and angiogenesis via time-resolved imaging data," *Cancers*, vol. 13, no. 12, 2021.
16. M. Fritz, P. K. **Jha**, T. Köppl, J. T. Oden, A. Wagner, and B. Wohlmuth, "Modeling and simulation of vascular tumors embedded in evolving capillary networks," *Computer Methods in Applied Mechanics and Engineering*, vol. 384, p. 113975, 2021.
17. M. Fritz, P. K. **Jha**, T. Köppl, J. T. Oden, and B. Wohlmuth, "Analysis of a new multispecies tumor growth model coupling 3D phase-fields with a 1D vascular network," *Nonlinear Analysis: Real World Applications*, vol. 61, p. 103331, 2021.
18. P. K. **Jha** and R. Lipton, "Finite element approximation of nonlocal dynamic fracture models," *Discrete & Continuous Dynamical Systems-B*, vol. 26, no. 3, p. 1675, 2021.
19. R. P. Lipton and P. K. **Jha**, "Nonlocal elastodynamics and fracture," *Nonlinear Differ. Equ. Appl. (NoDEA)*, vol. 28, no. 23, 2021.
20. P. K. **Jha**, P. S. Desai, D. Bhattacharya, and R. Lipton, "Peridynamics-based discrete element method (PeriDEM) model of granular systems involving breakage of arbitrarily shaped particles," *Journal of the Mechanics and Physics of Solids*, vol. 151, p. 104376, 2021.
21. P. K. **Jha**, L. Cao, and J. T. Oden, "Bayesian-based predictions of COVID-19 evolution in Texas using multispecies mixture-theoretic continuum models," *Computational Mechanics*, vol. 66, no. 5, pp. 1055–1068, 2020.
22. P. K. **Jha** and R. P. Lipton, "Kinetic relations and local energy balance for LEFM from a nonlocal peridynamic model," *International Journal of Fracture*, vol. 226, no. 1, pp. 81–95, 2020.
23. P. K. **Jha** and R. Lipton, "Finite element convergence for state-based peridynamic fracture models," *Communications on Applied Mathematics and Computation*, vol. 2, no. 1, pp. 93–128, 2020.
24. P. Diehl, P. K. **Jha**, H. Kaiser, R. Lipton, and M. Lévesque, "An asynchronous and task-based implementation of peridynamics utilizing HPX—the C++ standard library for parallelism and concurrency," *SN Applied Sciences*, vol. 2, no. 12, pp. 1–21, 2020.
25. P. K. **Jha** and R. Lipton, "Numerical convergence of finite difference approximations for state-based peridynamic fracture models," *Computer Methods in Applied Mechanics and Engineering*, vol. 351, pp. 184–225, 2019.
26. R. P. Lipton, R. B. Lehoucq, and P. K. **Jha**, "Complex fracture nucleation and evolution with nonlocal elastodynamics," *Journal of Peridynamics and Nonlocal Modeling*, vol. 1, no. 2, pp. 122–130, 2019.
27. R. Lipton, E. Said, and P. K. **Jha**, "Free damage propagation with memory," *Journal of Elasticity*, vol. 133, no. 2, pp. 129–153, 2018.
28. P. K. **Jha** and R. Lipton, "Numerical convergence of nonlinear nonlocal continuum models to local elastodynamics," *International Journal for Numerical Methods in Engineering*, vol. 114, no. 13, pp. 1389–1410, 2018.

29. P. K. **Jha** and R. Lipton, "Numerical analysis of nonlocal fracture models in Hölder space," *SIAM Journal on Numerical Analysis*, vol. 56, no. 2, pp. 906–941, 2018.

Accepted and Forthcoming

30. I. Galloway and P. K. **Jha**, "MatTO: A modular framework for material and topology optimization of stimulus-responsive soft materials," *Accepted for publication in Journal of Applied Mechanics*, 2026.
31. K. Enakoutsa, P. Bogard, and P. K. **Jha**, "Do attention maps in neural operators learn the Green's function?," *To appear in Mathematics and Mechanics of Complex Systems*, 2026.
32. P. K. **Jha** and K. Enakoutsa, "Neural operators with residual-based correction: Applications to optimization and Bayesian inference," *Accepted book chapter*, 2026.

Under Review and Preprints

33. P. K. **Jha**, K. Enakoutsa, I. Galloway, and H. Anderson, "Adapting neural operators for mechanics decisions under changing operating conditions," *Under review at CMAME; arXiv:2609.33978*, 2026.
34. I. Galloway and P. K. **Jha**, "Model-informed joint material-structural optimization of hard-magnetic soft materials," *Revised manuscript under review at JMPS; arXiv:2607.14397*, 2026.

Book Chapters

35. P. K. **Jha** and R. Lipton, "Well-posed nonlinear nonlocal fracture models associated with double-well potentials," in *Handbook of Nonlocal Continuum Mechanics for Materials and Structures* (G. Z. Voyiadjis, ed.), pp. 1417–1456, Cham: Springer International Publishing, 2019.
36. P. K. **Jha** and R. Lipton, "Finite differences and finite elements in nonlocal fracture modeling: A priori convergence rates," in *Handbook of Nonlocal Continuum Mechanics for Materials and Structures* (G. Z. Voyiadjis, ed.), pp. 1457–1494, Cham: Springer International Publishing, 2019.
37. R. Lipton, E. Said, and P. K. **Jha**, "Dynamic brittle fracture from nonlocal double-well potentials: A state-based model," in *Handbook of Nonlocal Continuum Mechanics for Materials and Structures* (G. Z. Voyiadjis, ed.), pp. 1265–1291, Cham: Springer International Publishing, 2019.
38. R. Lipton, E. Said, and P. K. **Jha**, "Dynamic damage propagation with memory: A state-based model," in *Handbook of Nonlocal Continuum Mechanics for Materials and Structures* (G. Z. Voyiadjis, ed.), pp. 1495–1523, Cham: Springer International Publishing, 2019.

Teaching Experience

South Dakota School of Mines & Technology

Fall 2026	ME 736 Advanced Finite Element Methods; ME 322 Machine Design I.
Spring 2026	ME 316 Solid Mechanics; ME 322 Machine Design I.
Fall 2025	ME 428/528 Applied Finite Element Analysis; ME 322 Machine Design I.
Fall 2024 & Spring 2025	ME 322 Machine Design I.

Prior Teaching

Spring 2024	Module coordinator and instructor, Engineering Principles (M21946) and Technology Concepts (M21967), University of Portsmouth.
Fall 2021	Instructor, Engineering Computation (COE 311K) and Numerical Methods in Biomedical Engineering (BME 313L), The University of Texas at Austin.

Course Development and Open Educational Resources

- **Advanced Finite Element Methods**: Developed a graduate course integrating variational foundations, error estimation and adaptivity, mixed and nonlinear methods, transient problems, and PDE-constrained optimization through Python/FEniCSx. Created an online book and public course repository. Fall 2026.

- **Machine Design I**: Redesigned the junior-level course around a reusable computational-design approach in which students formulate variables, constraints, and objectives; explore feasible designs in Python and MATLAB; select standard components; and verify designs using mechanics. Fall 2026.

Professional Activities

Conference Organization (Selected)

- Minisymposium on "Computational Modeling and Design Optimization of Functional Materials" at the 19th U. S. National Congress on Computational Mechanics, organized with Daicong Da and Yingqi Jia. Jul 2027.
- Minisymposium 207 on "Operator Learning in Mechanics with Robust Error Estimation, Control, and Generalization" at the 20th U. S. National Congress on Theoretical and Applied Mechanics. Jun 2026.
- Minisymposium 1301 on "Uncertainty Quantification and Scientific Machine Learning for Predictive Modeling and Decision-Making in Complex Systems" at the 18th U. S. National Congress on Computational Mechanics. Jul 2025.
- Minisymposium on "Integrating machine learning and numerical methods to accelerate engineering design" at 2nd IACM MMLDE-CSET. Sep 2023.
- Minisymposium M403 on "Uncertainty quantification for learning and data-driven predictive modeling of complex systems" at the 17th U. S. National Congress on Computational Mechanics. Jul 2023.
- Main organizer, USACM thematic conference on computational oncology. Jan 2022.
- Minisymposium M19 on "Nonlocal models in mathematics and computation" at the SIAM TX-LA 3rd Annual Meeting. Oct 2020.

Mentoring

- Advising three graduate researchers in mechanics, computational design, and scientific machine learning in the CEAD Lab. Current work includes a computational-twin manuscript with Ian Galloway and Henry Anderson under review at CMAME and research by Henry Anderson and Caleb Oien on particle–matrix damage, fatigue, and competing fracture paths in composites. Fall 2024–present.
- Mentored Ian Galloway from undergraduate research into graduate study in joint material–structural optimization of magnetic soft materials. He presented this work at USNCCM 18 (Jul 2025) and USNCTAM 20 (Jun 2026) and is first author of an article accepted by JAM and a revised manuscript under review at JMPS.
- Co-mentored a Google Summer of Code student developing computational tools for nonlocal diffusion. Repository. Summer 2020.

Open-Source Software

PeriDEM (Jha, JOSS 2025; Jha et al., JMPS 2021); **neural_operators** (Jha, Mathematics 2026); **MatTO** (Galloway & Jha, accepted in JAM, 2026); **NLMech** (Jha & Diehl, JOSS 2021); **finite-element-methods** (online book, under development).

Service

- Member, Faculty Senate, South Dakota Mines. Fall 2025 – present.
- Advisor, India Club, South Dakota Mines. Fall 2025 – present.

Editorial and Review Service

- Associate Editor, *Journal of Peridynamics and Nonlocal Modeling*; Topic Editor, *Journal of Open Source Software*; Editorial Board Member, *Scientific Reports*.
- Guest Editor, *Computing in Science & Engineering* special issue "Celebrating the Life and Work of J. Tinsley Oden," Apr 2026.
- NSF CMMI panel reviewer: two panels in 2025 and one panel in 2026; NSF ad hoc reviewer, 2025.
- Reviewer for CMAME (30+ reviews), JMPS, SINUM, M3AS, MMS, Mathematical Reviews, and JAM.

Selected Invited Seminars and Colloquia

1. *Mechanics of field-responsive materials and reliable data-driven prediction.* Departmental Seminar, Department of Mechanical Engineering, Indian Institute of Science. Oct 2026.
2. *Error correction and adaptation for reliable neural operators.* Department of Computational and Data Sciences, Indian Institute of Science. Oct 2026.
3. *Rethinking reliability in neural operators through residual and agent-based corrections.* Center for Nonlinear Analysis Seminar, Carnegie Mellon University. Oct 2025.
4. *Residual and agent-based corrections and beyond for reliable neural operators.* Applied Analysis and Data-Driven Mathematical Modeling Seminar, University of Nebraska–Lincoln. Oct 2025.
5. *Rethinking reliability in neural operators through residual and agent-based corrections.* Mathematics Colloquium, University of Nebraska–Lincoln. Oct 2025.
6. *Peridynamics for solids and granular media: From fracture to computational frameworks.* Mechanical Engineering Seminar, University of Wyoming. Sep 2025.
7. *Corrector operator to enhance accuracy and reliability of neural-operator surrogates of nonlinear variational boundary-value problems.* CRUNCH Seminar, Brown University. Aug 2023.
8. *High-fidelity mechanistic modeling of tumor growth at the tissue scale.* Babuška Forum, Oden Institute, The University of Texas at Austin. Sep 2021.
9. *Modeling failure in solids and tissue-scale tumor growth via high-fidelity computational methodologies.* Department of Computational and Data Science Seminar, Indian Institute of Science. May 2021.

Selected Conference and Workshop Presentations

1. *Rethinking reliability of neural operators via residual and agent-based error correction.* USNCTAM 20, Pasadena. Jun 2026.
2. *Robust computational design of magnetic soft structures under model dependence.* Presented by CEAD Lab student Ian Galloway, USNCTAM 20, Pasadena. Jun 2026.
3. *Seamless multiphysics coupling with peridynamics enabled by nodal finite element approximation.* USNCCM 18, Chicago. Jul 2025.
4. *Reliable neural operators: Error control through residual correction and beyond.* Accuracy and Efficiency in Scientific Machine Learning Workshop, IVADO/CRM, Montreal. Jun 2025.
5. *Application of peridynamics to granular media.* Engineering Mechanics Institute Conference, Anaheim. May 2025.
6. *Goal-oriented a-posteriori estimation of model error as an aid to parameter estimation.* USNCCM 17, Albuquerque. Jul 2023.

References

Prof. Kaushik Dayal
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Carnegie Mellon University, Pittsburgh, PA 15213, USA
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P: 412-268-2949 W: Homepage

Prof. Robert Lipton
Department of Mathematics
Louisiana State University, Baton Rouge, LA 70803, USA
E: lipton@lsu.edu
P: 225-578-1569 W: Homepage

Prof. David T. Fuentes
Department of Imaging Physics
MD Anderson Cancer Center, Houston, TX 77030, USA
E: DTFuentes@mdanderson.org
P: 713-745-3377 W: Homepage